

Iterative Thresholding

Consider the below image with pixel intensities. Apply iterative thresholding to find the optimal threshold for this image.

100	102	110
120	125	130
135	140	145

① Calculate Initial threshold (T_0):

Start with mean of all pixel intensities.

$$T_0 = \frac{100 + 102 + 110 + 120 + 125 + 130 + 135 + 140 + 145}{9} \\ = \frac{1107}{9} \approx 123$$

② Iteration 1:

(a) Separate Pixels:

Pixels with values below 123 $\rightarrow$ 100, 102, 110, 120

Pixels with values above or equal to 123 $\rightarrow$ 125, 130, 135, 140, 145

(b) Calculate Group Means:

(i) Mean of pixels below 123:

$$\text{Mean}_{\text{low}} = \frac{100 + 102 + 110 + 120}{4} = \frac{432}{4} = 108$$

(ii) Mean of pixels above 123:

$$\text{Mean}_{\text{high}} = \frac{125 + 130 + 135 + 140 + 145}{5} = \frac{675}{5} = 135$$

(c) Calculate new threshold:

$$T_1 = \frac{\text{Mean}_{\text{low}} + \text{Mean}_{\text{high}}}{2} = \frac{108 + 135}{2} = 121.5 \approx 122$$

③ Iteration 2:

(a) Separate Pixels:

(i) Pixels with values below 122 : 100, 102, 110, 120

(ii) Pixels with values above or equal to 122 : 125, 130, 135, 140, 145

(b) Calculate Group Means:

(i) Mean of pixels below 122:

$$\text{Mean}_{\text{low}} = \frac{100 + 102 + 110 + 120}{4} = \frac{432}{4} = 108$$

(ii) Mean of pixels above 122:

$$\text{Mean}_{\text{high}} = \frac{125 + 130 + 135 + 140 + 145}{5} = \frac{675}{5} = 135$$

(c) Calculate new threshold:

$$T_2 = \frac{108 + 135}{2} = 121.5 \approx 122$$

Since, the threshold hasn't changed from previous iteration, the process converges at 122.

④ Final Segmentation:

With a threshold of 122:

(i) Pixels below 122 (background) would be 100, 102, 110, 120.

(ii) Pixels above or equal to 122 (foreground) would be 125, 130, 135, 140, 145.

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 \end{bmatrix}$$